

Vehicle Mileage and Reservation Tracker

Portfolio overview | Excel/VBA operational workflow

Independently designed and built end to end by Charles Monsanto, from requirements and control design through implementation, testing, privacy sanitization, and release documentation.

PURPOSE

A practical desktop application that brings mileage entry, vehicle reservations, maintenance status, corrections, and recurring reporting into one governed workflow.

PROBLEM

Separate logs, calendars, and messages make it harder to prevent conflicts, preserve dated status, correct records safely, and keep management views consistent.

SOLUTION AND CONTROLS

The dashboard validates transactions before table writes, blocks invalid reservations, supports grouped corrections, and refreshes the affected reports after approved actions.

EXECUTIVE VALUE

Reviewers can see product ownership, process analysis, internal-control thinking, application engineering, reporting design, and disciplined release preparation in one finished artifact.

VEHICLE MILEAGE AND RESERVATION TRACKER

LOG MILEAGE

Trip Date

7/20/2026

CAL

Vehicle

Odo Start

Odo End

SAVE MILEAGE

CLEAR

GO TO MILEAGE TRACKER TAB

RESERVE VEHICLE

Date

7/20/2026

CAL

Vehicle

Staff

Start

8:00 AM

End

5:00 PM

Purpose

SAVE RESERVATION

CLEAR

SHOP STATUS

Vehicle

Location

Shop - Central Garage

MARK IN SHOP

MARK AVAILABLE

RESERVATIONS

Open Reservations tab to manage weekly bookings.

GO TO RESERVATIONS TAB

STATUS LEGEND:

Reserved

Available

In Shop

MONTHLY MILEAGE REVIEW

Fiscal Year

FY 2026-27

Report Month

July 2026

Vehicle	Last Mileage	Checkout Status (This Week)	Month Miles	Trip Days	Trips	Scheduled Hours	FYTD Miles
Demo SUV	52,530	Available	530	5	5	16.0	530
Demo Crossover	41,925	Shop - Central Garage	225	2	2	0.0	225
Demo Compact	28,740	7/20 Reserved - Cameron Davis	440	4	4	32.0	440
Demo Pickup	63,110	Available	310	3	3	16.0	310
TOTAL			1,505	14	14	64.0	1,505

Note: Please ensure all mileage and checkout information is accurate before saving.

CLEAR FIELDS

EDIT MILEAGE

EDIT RESERVATION

Synthetic dashboard showing controlled entry, maintenance, correction, and reporting workflows.

Executive evidence

The application is usable, reviewable, and documented. It combines controlled transactions, traceable corrections, dated maintenance history, and recurring management views without requiring a server, add-in, or external data connection.

Capabilities and Review Evidence

Workflow controls, reporting, lightweight implementation, and release verification

WEEKLY RESERVATIONS									
Select week									
Month		July		Year		2026		Month	
Date		7/20/2026		CAL		PREVIOUS WEEK		NEXT WEEK	
Showing July 19 - July 25, 2026									
DASHBOARD									

DEMO FLEET					
SUN	19-Jul	DEMO SUV	DEMO CROSSOVER	DEMO COMPACT	DEMO PICKUP
DATE	DAY	DEMO SUV	DEMO CROSSOVER	DEMO COMPACT	DEMO PICKUP
		Plate: DEMO-01	Plate: DEMO-02	Plate: DEMO-03	Plate: DEMO-04
		1001001	1001002	1001003	1001004
SUN	7/19		Shop - Central Garage		
MON	7/20		Shop - Central Garage	8:00 AM-5:00 PM Cameron Davis Project coordination	
TUE	7/21		Shop - Central Garage		
WED	7/22		Shop - Central Garage		
THU	7/23		Shop - Central Garage		
FRI	7/24		Shop - Central Garage		
SAT	7/25		Shop - Central Garage		

Weekly reservations and dated availability

CAPABILITIES

- Mileage entry with odometer validation and recalculation.
- Single- and multi-day reservations, with multiple non-overlapping same-day bookings.
- Dated maintenance history blocks affected reservations and remains visible after return.
- Weekly schedules, monthly mileage, trip counts, scheduled hours, fiscal-year totals, and status.

LIGHTWEIGHT BY DESIGN

Structured Excel tables hold local data; thin worksheet events dispatch actions to standard VBA modules. The complete plain-text source is available for review, with no server, external database, API, add-in, or data connection.

Testing, privacy, and evidence boundaries

Regression checks cover startup, save/edit/delete, recalculation, boundary conflicts, grouped corrections, maintenance, reports, package integrity, and Excel state restoration. Test-machine measures: functional open 1.844 s; dashboard refresh 0.219 s. All records are synthetic; no organization data or external connections are included. No measured business time or cost savings are claimed.

<https://github.com/cbmons4/vehicle-mileage-reservation-tracker>

Vehicles	Jul-26	Demo SUV (1001001)	Demo Crossover (1001002)	Demo Compact (1001003)	Demo Pickup (1001004)
Day	Date	Trip(s)	End Mileage	Trip(s)	End Mileage
Wednesday	7/1/2026			1	41,825
Thursday	7/2/2026	1	52,125		
Friday	7/3/2026				1 28,410
Saturday	7/4/2026				
Sunday	7/5/2026				
Monday	7/6/2026	1	52,210		
Tuesday	7/7/2026				1 62,920
Wednesday	7/8/2026			1	28,520
Thursday	7/9/2026		1	41,925	
Friday	7/10/2026	1	52,340		1 28,630
Saturday	7/11/2026				
Sunday	7/12/2026				
Monday	7/13/2026				1 63,020
Tuesday	7/14/2026				
Wednesday	7/15/2026	1	52,420		
Thursday	7/16/2026				1 63,110
Friday	7/17/2026				
Saturday	7/18/2026		IN SHOP	Central Garage	
Sunday	7/19/2026		IN SHOP	Central Garage	
Monday	7/20/2026	1	52,530	IN SHOP	Central Garage 1 28,740
Tuesday	7/21/2026			IN SHOP	Central Garage
Wednesday	7/22/2026			IN SHOP	Central Garage
Thursday	7/23/2026			IN SHOP	Central Garage
Friday	7/24/2026			IN SHOP	Central Garage
Saturday	7/25/2026			IN SHOP	Central Garage
Sunday	7/26/2026			IN SHOP	Central Garage
Monday	7/27/2026			IN SHOP	Central Garage
Tuesday	7/28/2026			IN SHOP	Central Garage
Wednesday	7/29/2026			IN SHOP	Central Garage
Thursday	7/30/2026			IN SHOP	Central Garage
Friday	7/31/2026			IN SHOP	Central Garage
July 2026 TOTALS		5	530	2	225
CUMULATIVE MILEAGE		Ending Mileage	52,530	Ending Mileage	41,925
				Ending Mileage	28,740
				Ending Mileage	63,110

Monthly mileage history and maintenance status

WORKFLOW CONTROLS

- Required-field, date, time, vehicle, driver, and odometer checks before table changes.
- Half-open [start, end) logic allows adjacent bookings and blocks actual overlaps.
- Recent-entry actions keep multi-day reservations grouped and recalculate affected outputs.
- Generated sheets are protected while approved dashboard actions remain available.